

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

A COMPREHENSIVE COVERAGE

MODULE - 25

WATER POLLUTION -

CORAL REEFS

- Coral reefs are colonies of individual animals called polyps which are related to jellyfish. A single coral polyp may be as large as a saucer or smaller than the head of a pin.
- Despite the fact that coral polyps look more like rocks or plants, they are animals.
- Coral reefs are called the 'rainforests of oceans'.
- They occupy less than one percent of the ocean floor, yet are home to more than a quarter of all marine species: crustaceans, reptiles, seaweeds, bacteria, fungi, and over 4000 species of fish make their home in coral reefs.





TYPES OF CORALS

There are two types of Corals

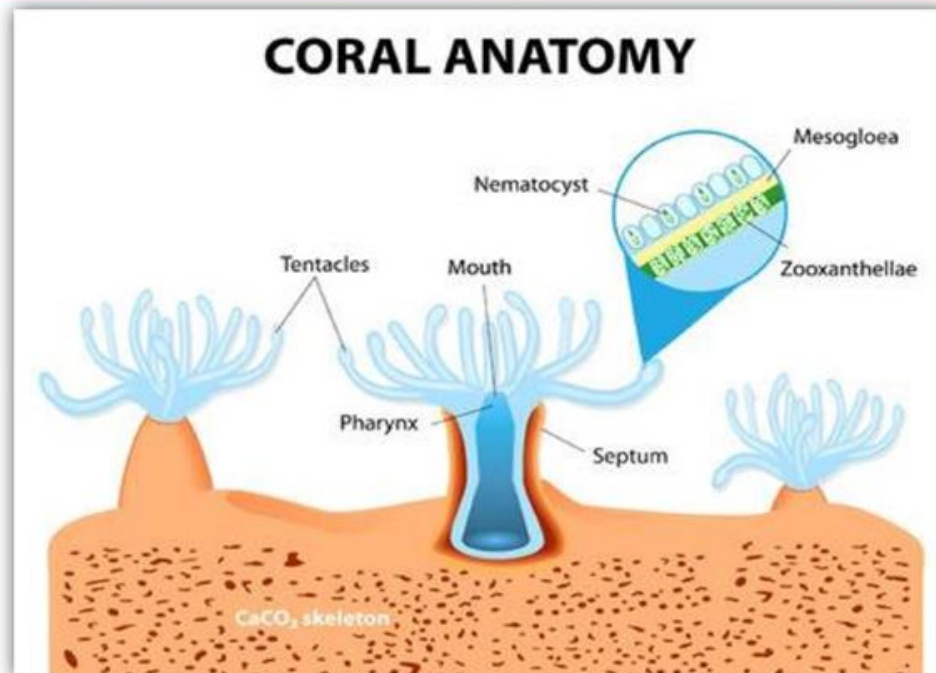
- ✓ Hard Corals (also called stony corals) and
- ✓ Soft Corals



TYPES OF CORALS

1 HARD CORALS

- Hard corals, also called reef-building corals, produce a rock-like skeleton made of calcium carbonate. These skeletons form reef.
- Hard corals rely on symbiotic algae (zooxanthellae) living within their tissues for nutrition and energy to build their skeleton. They must therefore live in shallow clear water to allow sunlight to reach the algae.



TYPES OF CORALS

1 HARD CORALS

- The number of coral species in each reef varies.
- The Great Barrier Reef off Australia has over 600 species of coral while a Caribbean reef has about 65.
- Today many reefs have 40 to 50 percent less coral than they did just 30 years ago.



CORAL REEFS

IDEAL CONDITIONS FOR CORAL GROWTH

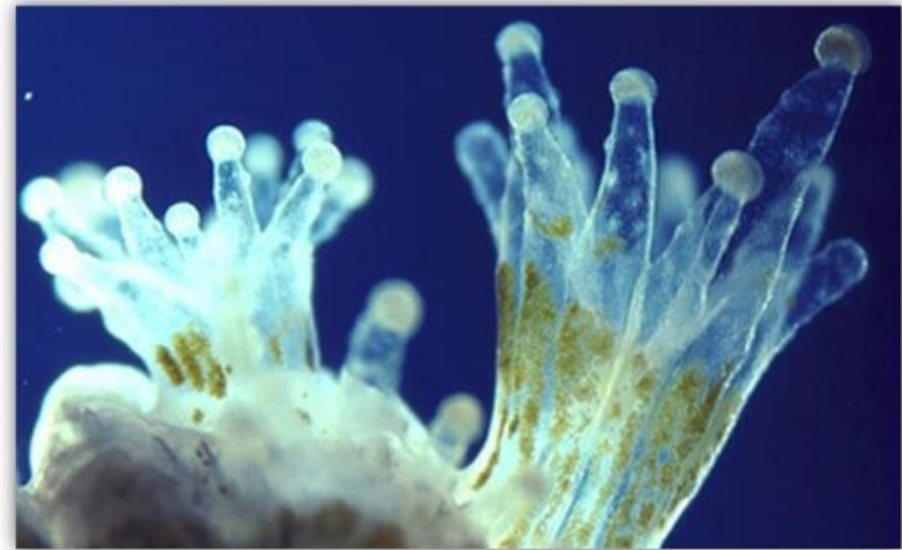
- **Stable climatic conditions** - Corals are highly susceptible to quick changes. They grow in regions where climate is significantly stable for a long period of time.
- **Warm waters** - Corals thrive in tropical waters (30°N and 30°S latitudes). The temperature of water should be around 25°C and diurnal and annual temperature ranges should be very narrow.
- **Shallow water** - Corals require good amount of sunlight to survive. So, they are found at shallow depths where there is abundant sunlight available.
- **Clear salt water** - Clear salt water is suitable for coral growth, while both fresh water and highly saline water are harmful.
- **No pollution** - Corals are highly fragile and are vulnerable to pollution and even a minute increase in marine pollution can be catastrophic.



CORAL REEFS

CORALS AND ZOOXANTHELLAE

- Coral Polyps have a symbiotic relationship with a species of single-celled microscopic algae known as Z zooxanthellae.
- Zooxanthellae are autotrophic; they prepare their own food through photosynthesis.
- Zooxanthellae lives within the tissues of coral polyp and carries out photosynthesis and provides the coral polyp with food and energy.
- The host coral polyp in return provides the zooxanthellae with a protected environment to live within and a continuous supply of CO_2 to carry out photosynthesis.



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